

On the C_4 -destroying of Bipartite Permutation Graphs

(Extended Abstract)

*Sheng-Lung Peng, Jian-Hong Lee, Jun-Rong Lin, Chih-Shan Liu
Department of Computer Science and Information Engineering
National Dong Hwa University
Hualien, Taiwan, R.O.C.

Abstract

In this paper we propose a succinct representation for bipartite permutation graphs. Using this representation, we propose a unified approach on a given bipartite permutation graph G for the following problems: (1) adding the minimum number of edges on G such that the resulting graph has no C_4 (the minimum fill-in problem); (2) adding edges on G such that the resulting graph has no C_4 and its maximum clique size is as small as possible (the treewidth problem).

1 Introduction

Let $G = (V, E)$ be a finite, simple, and undirected graph where V and E denote the vertex and edge sets of G , respectively. For a vertex set $W \subseteq V$, let $G(W)$ denote the induced subgraph of G which is induced by W . A *biclique* is a complete bipartite graph. Let $N(v) = \{w | (v, w) \in E\}$ and $N[v] = N(v) \cup \{v\}$.

A graph $G = (V, E)$ is *bipartite* if V can be partitioned into two sets S and T such that every edge has its ends in different sets: vertices in the same set must not be adjacent. Equivalently, G is bipartite if and only if it is 2-colorable and is denoted as $G = (S, T, E)$.

A graph G is a *permutation graph* if there is some pair P, Q of permutations of the vertex set such that there is an edge between vertices x and y if and only if x precedes y in one of P, Q , while y precedes x in the other. A permutation graph can also be described by a permutation diagram. Let the vertices in the order of P be on a line and the vertices in the order of Q be on a parallel line. Draw an edge between two vertices that having the same assigned number. Then, there is an edge

(x, y) in G if and only if the line segment for x intersects the line segment for y .

Bipartite permutation graphs simultaneously have both of the properties of bipartite graphs and permutation graphs. Bipartite permutation graphs are first proposed by Spinrad, Brandstadt, and Stewart in 1985 [19].

A graph is *chordal* if it does not contain a chordless cycle of length greater than three. The *treewidth* (respectively, *minimum fill-in*) of a graph is the smallest maximum clique size (respectively, the minimum number of additional edges) of its all possible chordal supergraphs. The *treewidth problem* (respectively, *minimum fill-in problem*) of a graph is to find its treewidth (respectively, minimum fill-in).

The treewidth problem is NP-complete on cobipartite graphs (complement of bipartite graphs) [1], bipartite graphs [15], and graphs of maximum degree at most 9 [6]. For some special classes of graphs, it has been shown that the treewidth can be computed in polynomial time, such as cographs [5], circular-arc graphs [20], chordal bipartite graphs [16], permutation graphs [3], circle graphs [13], d -trapezoid graphs [4], and distance hereditary graphs [8].

The minimum fill-in problem is NP-complete on cobipartite graphs [22] and bipartite graphs [21]. The known graph classes for which the minimum fill-in can be computed by a polynomial time algorithm are cographs [10], d -trapezoid graphs [4], bipartite permutation graphs [19], chordal bipartite graphs [14, 9], and circle and circular-arc graphs [18].

In this paper, we first propose a representation for bipartite permutation graphs. That is, we show that a bipartite permutation graph can be obtained from a sequence of bicliques, e.g., $B_1, B_2, \dots, B_r$. Using this structure, we show that the treewidth and minimum fill-in problems can

*Corresponding author: lung@csie.ndhu.edu.tw

be solved in $O(r^2)$ time on bipartite permutation graphs. In the case of $r = o(n)$, it is an improvement where n is the number of vertices in the input graph.

The rest of this paper is organized as follows. In Section 2, we give a new representation for bipartite permutation graphs. Our unified algorithm for minimum fill-in and treewidth problems on bipartite permutation graphs is shown in Section 3. Finally, we give the concluding remarks in the last section.

2 Biclique structure of bipartite permutation graphs

Let $G = (S, T, E)$ be a bipartite graph. An ordering of the vertices in S has the *adjacency property* if for each vertex x in T , $N(x)$ consists of which are consecutive in the ordering of S . An ordering of the vertices in S has the *enclosure property* if for every pair of vertices v, w in T such that $N(v)$ is a subset of $N(w)$, vertices in $N(w) \setminus N(v)$ occur consecutively in the ordering of S . A *strong ordering* of the vertices of $G = (S, T, E)$ consists of an ordering of S and an ordering of T such that for all $(s, t), (s', t')$ in E where s, s' are in S and t, t' are in T , $s < s'$ and $t > t'$ imply (s, t') and (s', t) are in E . In [19], Spinrad et al. give the following characterization of a bipartite permutation graph.

Theorem 2.1 [19] The following statements are equivalent:

1. $G = (S, T, E)$ is a bipartite permutation graph.
2. There is a strong ordering of $S \cup T$.
3. There exists an ordering of S which has the adjacency and enclosure properties.

For a biclique $B = (X, Y, E')$, $E' = \{(x, y) \mid x \in X \text{ and } y \in Y\}$ is completely determined by X and Y . Thus, in the following, we use $B = (X, Y)$ to denote a biclique. Let $first(X, k)$ (respectively, $last(X, k)$) be the first (respectively, last) k vertices of X with ordering. For example, assume that $X = \{x_1, x_2, \dots, x_r\}$. Then $first(X, k) = \{x_1, x_2, \dots, x_k\}$ and $last(X, k) = \{x_{r-k+1}, x_{r-k+2}, \dots, x_{r-1}, x_r\}$.

For two bicliques $B_1 = (X_1, Y_1)$ and $B_2 = (X_2, Y_2)$ and two positive integers k and l , we define a concatenating operation $\oplus$ on B_1 and B_2 as follows:

Let $G = (S, T, E) = B_1 \oplus_l^k B_2$ where $S = X_1 \cup X_2$ by letting $last(X_1, k) = first(X_2, k)$ and $T = Y_1 \cup Y_2$ by letting $last(Y_1, l) = first(Y_2, l)$ with E being the union of edges in B_1 and B_2 and k, l satisfying the following two conditions.

1. $k + l \geq 1$,
2. neither $|X_1| = |X_2| = k$ nor $|Y_1| = |Y_2| = l$.

By Condition 1, G is connected. By Condition 2, it is impossible to have $|X_1| = |X_2| = k$ (or $|Y_1| = |Y_2| = l$). Otherwise, $B_1 \oplus_l^k B_2$ becomes a biclique. Note that B_1 and B_2 are two maximal bicliques of G . For convenience, we say that $last(X_1, k) \cup last(Y_1, l) = first(X_2, k) \cup first(Y_2, l)$ is the intersection of B_1 and B_2 .

For convenience, if there is no confusion, then $B_1 \oplus_l^k B_2$ is briefed as $B_1 \oplus B_2$. For a sequence of bicliques $B_1, B_2, \dots, B_r$, we define G_i as follows:

- (1) $G_1 = B_1$,
- (2) let $G_i = B_1 \oplus B_2 \oplus \dots \oplus B_i$ be the graph by concatenating $B_1, B_2, \dots, B_i$,
- (3) $G_{i+1} = G_i \oplus B_{i+1}$ where $\oplus$ is applied on B_i and B_{i+1} .

Lemma 2.2 G_r is a bipartite permutation graph.

Proof. We prove this lemma by induction.

- (i) G_1 is a biclique which is a bipartite permutation graph.
- (ii) Assume that $G_i = B_1 \oplus B_2 \oplus \dots \oplus B_i$ is a bipartite permutation graph. Now, we consider $G_{i+1} = G_i \oplus_l^k B_{i+1} = B_1 \oplus B_2 \oplus \dots \oplus B_i \oplus_l^k B_{i+1}$ where $B_i = (X_i, Y_i)$ and $B_{i+1} = (X_{i+1}, Y_{i+1})$. It is not hard to see that the neighborhood is changed only in the vertices of $B_i \cap B_{i+1}$. By the definition, vertices in $B_i \cap B_{i+1}$ satisfy the adjacency property. It is also not hard to see that they satisfy the enclosure property. That is, G_{i+1} is a bipartite permutation graph by Theorem 2.1.

Thus, by induction, G_r is a bipartite permutation graph. $\square$

If G is obtained by $B_1 \oplus_{l_1}^{k_1} B_2 \oplus_{l_2}^{k_2} \dots \oplus_{l_{r-1}}^{k_{r-1}} B_r$, then we say that $B_1, B_2, \dots, B_r$ is a *generating base* of G with respect to $(\oplus_{l_1}^{k_1}, \oplus_{l_2}^{k_2}, \dots, \oplus_{l_{r-1}}^{k_{r-1}})$. On the other hand, given a bipartite permutation graph $G = (S, T, E)$ with a strong ordering, we can obtain a generating

base as follows. Assume that $S = \{s_1, s_2, \dots, s_a\}$ and $T = \{t_1, \dots, t_b\}$. Let the largest index in $N(s_1)$ (respectively, $N(t_1)$) be j (respectively, i). Let $X_1 = \{s_1, \dots, s_j\}$ and $Y_1 = \{t_1, \dots, t_i\}$. Then $B_1 = (X_1, Y_1)$ is a maximal biclique of G . After B_1 is determined, we delete s_1 (respectively, t_1) and the other vertices whose neighborhood is $N(s_1)$ (respectively, $N(t_1)$). It is not hard to see the remaining graph is still a bipartite permutation graph and the ordering of vertices is still strong. By repeating this process, we can obtain $B_2, B_3, \dots, B_r$.

Finally, we have the following theorem.

Theorem 2.3 *G is a bipartite permutation graph if and only if there exists a generating base for G .*

3 A unified algorithm

Given a bipartite permutation graph $G = (S, T, E)$ with vertices of S and T in strong ordering, each crossing of (u, y) and (w, x) is contained in a cycle u, y, w, x, u . That is, each crossing pair represents a 4-cycle in G . By the definition, a graph is chordal if every cycle of length greater than three has a chord. Therefore, to form a chordal graph, every 4-cycle must be destroyed (by adding edges). The set of all the additional edges is called C_4 -destroying set. A previous investigation by Spinrad et al. [19] showed that the union of G and any minimal C_4 -destroying set is chordal. For convenience, the union is called a *triangulation* of G . We denote that E' is a minimal C_4 -destroying set and $G' = (S \cup T, E \cup E')$ is a chordal graph. For a vertex set W , let $E_W = \{(u, v) \in E \mid u, v \in W\}$. Spinrad et al. showed the following lemma.

Lemma 3.1 [19] *If (u, w) is in E'_S (E'_T), then for all v in S (T) such that $u < v < w$, the edges (u, v) and (v, w) are also in E'_S (E'_T).*

By Lemma 3.1 we have the following lemma.

Lemma 3.2 *Let $B = (X, Y)$ be a biclique in bipartite permutation graph G , then either X or Y must form a clique in any C_4 -destroying set of G .*

Intuitively, Lemma 3.2 tells us that we need to find all the possible maximal bicliques of G for C_4 -destroying. Let $G = (S, T, E)$ be a bipartite permutation graph with its generating base $\{B_1, \dots, B_r\}$ where $B_i = (X_i, Y_i)$ for all $i \in \{1, \dots, r\}$. Assume that $T(G_{i-1})$ is a triangulation of G_{i-1} . Consider G_i and we define six operations for triangulating G_i as follows:

- OU_i^1 : in the case that Y_{i-1} is a clique in $T(G_{i-1})$, vertices in X_i form a clique;
- OU_i^2 : in the case that X_{i-1} is a clique in $T(G_{i-1})$, vertices in $X_j \cup X_i$ form a clique where j is the smallest index such that $X_j \cap X_i \neq \emptyset$;
- OU_i^3 : in the case that X_{i-1} is a clique in $T(G_{i-1})$, vertices in X_i and $Y_{i-1} \cap Y_i$ form respective cliques;
- OD_i^1 : in the case that X_{i-1} is a clique in $T(G_{i-1})$, vertices in Y_i form a clique;
- OD_i^2 : in the case that Y_{i-1} is a clique in $T(G_{i-1})$, vertices in $Y_j \cup Y_i$ form a clique where j is the smallest index such that $Y_j \cap Y_i \neq \emptyset$;
- OD_i^3 : in the case that Y_{i-1} is a clique in $T(G_{i-1})$, vertices in Y_i and $X_{i-1} \cap X_i$ form respective cliques.

It is not hard to see that if G_{i-1} is triangulated, then G_i can be triangulated by applying any operation defined above. If G_i is triangulated by OX_i^j , we call it Type Z_i^j of triangulation where $Z \in \{U, D\}$ and $j \in \{1, 2, 3\}$.

Lemma 3.3 *If there is a way to triangulate G_{i-1} , then G_i can be triangulated using each of the six operators.*

Proof. Let $B_{j \cap i}^{j \cup i}$ (respectively, $B_{j \cup i}^{j \cap i}$) be the biclique $B = (X_j \cup X_i, Y_j \cap Y_i)$ (respectively, $B = (X_j \cap X_i, Y_j \cup Y_i)$) with $Y_j \cap Y_i \neq \emptyset$ (respectively, $X_j \cap X_i \neq \emptyset$). By the concatenating operator, the maximal bicliques occurred in G_i but not in G_{i-1} are B_i , $B_{p \cap i}^{p \cup i}$, and $B_{q \cup i}^{q \cap i}$ for all possible p and q .

For each maximal biclique, at least one partite must become a clique by Lemma 3.1. Assume that $B_{i-1} = K_q^p$ and $B_i = K_s^r$. Since G_{i-1} has been triangulated, we only consider the new bicliques after adding B_i . It is not hard to see that the new maximal bicliques are B_i , are triangulated by the definition. In the case of K_{q+s-l}^k (respectively, K_l^{p+r-k}), vertices in $S \cap K_{q+s-l}^k$ (respectively, $T \cap K_l^{p+r-k}$) are contained in $S \cap K_s^r$ (respectively, $T \cap K_q^p$). Thus K_{q+s-l}^k (respectively, K_l^{p+r-k}) is triangulated. Hence type OU_i^1 is a triangulation of G_i . The other cases are similar. $\square$

By the definition, we have the following theorem.

Theorem 3.4 *For $i \in \{1, \dots, r\}$, we have the following statements.*

1. $U_i^1 = \min\{D_{i-1}^1 + OU_i^1, D_{i-1}^2 + OU_i^1, D_{i-1}^3 + OU_i^1\};$
2. $U_i^2 = \min\{U_{i-1}^1 + OU_i^2, U_{i-1}^2 + OU_i^2, U_{i-1}^3 + OU_i^2\};$
3. $U_i^3 = \min\{U_{i-1}^1 + OU_i^1, U_{i-1}^2 + OU_i^1, U_{i-1}^3 + OU_i^1\};$
4. $D_i^1 = \min\{U_{i-1}^1 + OD_i^1, U_{i-1}^2 + OD_i^1, U_{i-1}^3 + OD_i^1\};$
5. $D_i^2 = \min\{D_{i-1}^1 + OD_i^2, D_{i-1}^2 + OD_i^2, D_{i-1}^3 + OD_i^2\};$
6. $D_i^3 = \min\{D_{i-1}^1 + OD_i^1, D_{i-1}^2 + OD_i^1, D_{i-1}^3 + OD_i^1\}.$

Let $\text{mintri}(G_i)$ be the optimal triangulation of G_i . We have the following theorem.

Theorem 3.5 $\text{mintri}(G_i) = \min\{U_i^1, U_i^2, U_i^3, D_i^1, D_i^2, D_i^3\}.$

It is not hard to see that $U_1^1 = U_1^2 = U_1^3$ and $D_1^1 = D_1^2 = D_1^3$. We propose a unified algorithm for treewidth and minimum fill-in problems on bipartite permutation graphs using dynamic programming approach in the following.

Algorithm: Triangulating;

Input: $G = B_1 \oplus_1 \cdots \oplus_{r-1} B_r$ where $r > 1$, $B_i = K_{q_i}^{p_i}$ and $\oplus_i = \oplus_{l_i}^{k_i}$ for all i ;

Output: The optimal value Opt ;

1. **initialize** $U_1^1, U_1^2, U_1^3, D_1^1, D_1^2, D_1^3$;
2. **for** $i = 2$ **to** r **do begin**
3. compute $U_i^1, U_i^2, U_i^3, D_i^1, D_i^2, D_i^3$;
4. **end**;
5. $Opt^+(G) = \min\{U_r^1, U_r^2, U_r^3\}$;
6. $Opt^-(G) = \min\{D_r^1, D_r^2, D_r^3\}$;
7. $Opt(G) = \min\{Opt^+(G_r), Opt^-(G_r)\}$;

For treewidth, the variables and formula are as follows.

1. $U_1^1 = U_1^2 = U_1^3 = p_1$;
2. $D_1^1 = D_1^2 = D_1^3 = q_1$;
3. $U_i^1 = \min\{\max\{D_{i-1}^1, p_i + l_{i-1} - 1, q_{i-1} + k_{i-1} - 1\}, \max\{D_{i-1}^2, p_i + l_{i-1} - 1, q_{i-1} + k_{i-1} - 1\}, \max\{D_{i-1}^3, p_i + l_{i-1} - 1, q_{i-1} + k_{i-1} - 1\}\};$

4. $U_i^2 = \min\{\max\{U_{i-1}^1, p_i + \sum_{t=j}^{i-1} (p_t - k_t) - 1\}, \max\{U_{i-1}^2, p_i + \sum_{t=j}^{i-1} (p_t - k_t) - 1\}, \max\{U_{i-1}^3, p_i + \sum_{t=j}^{i-1} (p_t - k_t) - 1\}\};$
5. $U_i^3 = \min\{\max\{U_{i-1}^1, p_i + l_{i-1} - 1, p_{i-1} + l_{i-1} - 1\}, \max\{U_{i-1}^2, p_i + l_{i-1} - 1, p_{i-1} + l_{i-1} - 1\}, \max\{U_{i-1}^3, p_i + l_{i-1} - 1, p_{i-1} + l_{i-1} - 1\}\};$
6. $D_i^1 = \min\{\max\{U_{i-1}^1, q_i + k_{i-1} - 1, p_{i-1} + l_{i-1} - 1\}, \max\{U_{i-1}^2, q_i + k_{i-1} - 1, p_{i-1} + l_{i-1} - 1\}, \max\{U_{i-1}^3, q_i + k_{i-1} - 1, p_{i-1} + l_{i-1} - 1\}\};$
7. $D_i^2 = \min\{\max\{D_{i-1}^1, q_i + \sum_{t=j}^{i-1} (q_t - l_t) - 1\}, \max\{D_{i-1}^2, q_i + \sum_{t=j}^{i-1} (q_t - l_t) - 1\}, \max\{D_{i-1}^3, q_i + \sum_{t=j}^{i-1} (q_t - l_t) - 1\}\};$
8. $D_i^3 = \min\{\max\{D_{i-1}^1, q_i + k_{i-1} - 1, q_{i-1} + k_{i-1} - 1\}, \max\{D_{i-1}^2, q_i + k_{i-1} - 1, q_{i-1} + k_{i-1} - 1\}, \max\{D_{i-1}^3, q_i + k_{i-1} - 1, q_{i-1} + k_{i-1} - 1\}\};$

Similarly, the variables and formula for minimum fill-in can also be defined as a similar way. We omit the detail. Finally, we have the following theorem.

Theorem 3.6 *Algorithm Triangulating can be implemented in $O(r^2)$ time.*

4 Conclusion

Given a graph, computing the treewidth and minimum fill-in are interesting problems. The latter problem has an $O(n^5)$ -time algorithm on bipartite permutation graphs [19]. Both problems have been solved in $O(n^2)$ time on permutation graphs [3, 4]. In this paper, we propose a unified $O(r^2)$ -time algorithm for the treewidth and minimum fill-in problems on bipartite permutation graphs where r is the size of a generating base of the input graph. Note that r ranges from $O(1)$ to $O(n)$. In the case of $r = o(n)$, then our result is an improvement of others.

References

- [1] S. Arnborg, D. G. Corneil, and A. Proskurowski. Complexity of finding embeddings in a k -tree. *SIAM Journal of Algebraic and Discrete Methods*, 8:277–284, 1987.

- [2] H. L. Bodlaender. A tourist guide through treewidth. *Acta Cybernetica*, 11:1–23, 1993.
- [3] H. L. Bodlaender, T. Kloks, and D. Kratsch. Treewidth and pathwidth of permutation graphs. *SIAM Journal on Graph Discrete Mathematics*, 8:606–616, 1995.
- [4] H. L. Bodlaender, T. Kloks, D. Kratsch, and H. Müller. Treewidth and minimum fill-in on d -trapezoid graphs. *Journal of Graph Algorithms and Applications*, 2:1–23, 1998.
- [5] H. L. Bodlaender and R. H. Möhring. The pathwidth and treewidth of cographs. *SIAM Journal on Discrete Mathematics*, 6:181–188, 1993.
- [6] H. L. Bodlaender and D. M. Thilikos. Treewidth and small separators for graphs with small chordality. *Technical Report UU-CS-1995-02, Department of Computer Science, Utrecht University, Utrecht*, 1995.
- [7] A. Brandstadt, V. B. Le, and J. P. Spinrad, *Graph Classes: A Survey*, SIAM, 1999.
- [8] H. Broersma, E. Dahlhaus, and T. Kloks. A linear time algorithm for minimum fill in and treewidth for distance hereditary graphs. *Scientific program 5th Twente Workshop on Graphs and Combinatorial Optimization*, 48–50, 1997.
- [9] M. S. Chang. Algorithms for maximum matching and minimum fill-in on chordal bipartite graphs. *Proceedings of the 7th International Symposium Algorithms and Computation*, 146–155, 1996.
- [10] G. A. Dirac. Cographs: recognition, applications and algorithms. *Congressus Numerantium*, 43:249–258, 1984.
- [11] M. R. Garey and D. S. Johnson, *Computer and Intractability: A Guide to the Theory of NP-completeness*, W.H. Freeman, San Francisco, 1979.
- [12] M. C. Golumbic, *Algorithmic Graph Theory and Perfect Graphs*, Academic Press, New York, 1980.
- [13] T. Kloks. Treewidth of circle graphs. *Proceedings of the 4th International Symposium on Algorithms and Computation*, 108–117, 1993.
- [14] T. Kloks. Minimum fill-in for chordal bipartite graphs. *Technical Report RUU-CS-93-11, Department of Computer Science, Utrecht University, The Netherlands*, 1993.
- [15] T. Kloks. Treewidth—computations and approximations. *Lecture Notes in Computer Science* 842, 1994.
- [16] T. Kloks and D. Kratsch. Treewidth of chordal bipartite graphs. *Journal of Algorithms*, 19:266–281, 1995.
- [17] T. Kloks, D. Kratsch, and J. Spinrad. On treewidth and minimum fill-in of asteroidal triple-free graphs. *Journal of Theoretical Computer Science*, 175:309–335.
- [18] T. Kloks, D. Kratsch, and C. K. Wong. Minimum fill-in on circle and circular-arc graphs. *Proceedings of the 23 International Colloquium on Automata, Languages and Programming*, 256–267, 1996.
- [19] J. Spinrad, A. Brandstadt, and L. Stewart, Bipartite permutation graphs, *Discrete Applied Mathematics*, 18:279–292, 1987.
- [20] R. Sundaram, K. S. Singh, and C. P. Rangan. Treewidth of circular arc graphs. *SIAM Journal on Discrete Mathematics*, 7:647–655, 1994.
- [21] R. E. Tarjan. Decomposition by clique separators. *Discrete Mathematics*, 55:221–232, 1985.
- [22] M. Yannakakis. Computing the minimum fill-in is NP-complete. *SIAM Journal on Algebraic Discrete Methods*, 2:77–79, 1981.